

1 a.e. integrability

Theorem 1.1. Suppose $E \in \mathcal{F}$ is such that $\mu(E^c) = 0$ then,

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu,$$

provided either side exists. (*In fact, if one side exists, then so does the other, and they are equal*)

Proof. Step 1: Let $f \geq 0$ be measurable and suppose $\int_{\Omega} f \, d\mu < \infty$, Then

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu + \int_{E^c} f \, d\mu.$$

But

$$\int_{E^c} f \, d\mu = \int_{\Omega} f \mathbf{1}_{E^c} \, d\mu,$$

and $0 \leq f \mathbf{1}_{E^c} \leq \infty \cdot \mathbf{1}_{E^c}$, which by infinity arithmetic gives,

$$0 \leq \int_{\Omega} f \mathbf{1}_{E^c} \, d\mu \leq \infty \cdot \mu(E^c) = 0.$$

Hence $\int_{E^c} f \, d\mu = 0$, therefore

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu.$$

Step 2: For general f , write $f = f^+ - f^-$. Suppose first that

$$\int_{\Omega} f^+ \, d\mu < \infty.$$

Then, by Step 1,

$$\int_{\Omega} f^+ \, d\mu = \int_E f^+ \, d\mu.$$

Hence

$$\begin{aligned} \int_{\Omega} f \, d\mu &= \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu \\ &= \int_E f^+ \, d\mu - \int_E f^- \, d\mu \\ &= \int_E f \, d\mu. \end{aligned}$$

Now suppose $\int_{\Omega} f^- \, d\mu < \infty$, then the same argument gives,

$$\int_{\Omega} f \, d\mu = \int_E f \, d\mu.$$

Finally, if $\int_{\Omega} f^+ \, d\mu = \infty$ then,

$$\int_{\Omega} f \, d\mu = \int_{\Omega} f^+ \, d\mu - \int_{\Omega} f^- \, d\mu = +\infty$$

Whenever $\int_{\Omega} f^{-} d\mu < \infty$, by Step 1,

$$\int_E f^{+} d\mu = \infty, \quad \int_E f^{-} d\mu < \infty,$$

and hence

$$\boxed{\int_E f d\mu = +\infty}$$

The case for $\int_{\Omega} f^{-} d\mu = \infty$ is analogous. □

2 Change of Variables Formula

Suppose $(\Omega, \mathcal{F}, \mu) \xrightarrow{f} (\Omega', \mathcal{F}', \nu)$ and $\nu = \mu \circ f^{-1}$ where f is measurable. We want to write

$$\int_{\Omega'} g(\omega') d\nu(\omega')$$

in terms of the measure space $(\Omega, \mathcal{F}, \mu)$

3 Change of Variables Formula

Theorem 3.1 (Change of Variables Formula). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $(\Omega', \mathcal{F}')$ be a measurable space. Let $f : \Omega \rightarrow \Omega'$ be $\mathcal{F}/\mathcal{F}'$ -measurable. Let $\nu = \mu \circ f^{-1}$.

Then, for every non-negative measurable function $g : \Omega' \rightarrow [0, \infty]$,

$$\int_B g d\nu = \int_{f^{-1}(B)} g \circ f d\mu \quad (*) \quad \forall B \in \mathcal{F}'.$$

The same identity holds for every integrable measurable g .

Corollary 3.2. Suppose $f : \Omega \rightarrow \mathbb{R}$ is measurable, and let $\nu = \mu \circ f^{-1}$. Then

$$\int_{\Omega} f d\mu = \int_{\mathbb{R}} x \nu(dx).$$

Thus the integral of f can be written as an integral with respect to the distribution induced by f .

Proof. We prove the result by measure-theoretic induction on g .

Step 1: Suppose $g = \mathbf{1}_C$, and $C \in \mathcal{F}'$. Then,

$$\begin{aligned} \int_B g d\nu &= \nu(C \cap B) \\ &= \mu(f^{-1}(C \cap B)) \\ &= \mu(f^{-1}(C) \cap f^{-1}(B)) \\ &= \int_{f^{-1}(B)} \mathbf{1}_{f^{-1}(C)} d\mu = \int_{f^{-1}(B)} \mathbf{1}_C \circ f d\mu. \end{aligned}$$

Hence

$$\boxed{\int_B \mathbf{1}_C d\nu = \int_{f^{-1}(B)} \mathbf{1}_C \circ f d\mu.}$$

Step 2: If the result holds for $g_1, \dots, g_N \geq 0$ and $\alpha_1, \dots, \alpha_N > 0$, then, by linearity, $g = \sum_{i=1}^N \alpha_i g_i$ also satisfies $(*)$. Hence $(*)$ holds for every non-negative simple measurable function.

Step 3: Suppose $(*)$ holds for $g_n \geq 0$ and $g_n \uparrow g$. Then, by the Monotone Convergence Theorem,

$$\begin{aligned}\int_B g \, d\nu &= \lim_{n \rightarrow \infty} \int_B g_n \, d\nu \\ &= \lim_{n \rightarrow \infty} \int_{f^{-1}(B)} g_n \circ f \, d\mu \\ &= \int_{f^{-1}(B)} g \circ f \, d\mu.\end{aligned}$$

Since every non-negative measurable function is the pointwise limit of an increasing sequence of non-negative simple functions, $(*)$ holds for every $g \geq 0$ measurable.

Step 4: For a general measurable g , write $g = g^+ - g^-$. Applying the result separately to g^+ and g^- , whenever the corresponding integrals are finite, gives

$$\int_B g \, d\nu = \int_{f^{-1}(B)} g \circ f \, d\mu.$$

Hence the result follows. $\square$

4 Riemann Integral

Let $-\infty < a < b < \infty$ and let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$, i.e. $a = x_0 < x_1 < \dots < x_n = b$.

Define the norm (mesh) of P by

$$\|P\| = \max_{0 \leq i \leq n-1} (x_{i+1} - x_i).$$

We say P' is finer than P if $P \subseteq P'$. Then $\|P\| \geq \|P'\|$.

Define the *Riemann upper sum* as

$$U(f, P) = \sum_{i=0}^{n-1} \sup_{x \in [x_i, x_{i+1}]} f(x) (x_{i+1} - x_i),$$

and the *Riemann lower sum* as

$$L(f, P) = \sum_{i=0}^{n-1} \inf_{x \in [x_i, x_{i+1}]} f(x) (x_{i+1} - x_i).$$

By definition, $U(f, P) \geq L(f, P)$. If P' is finer than P , then

$$L(f, P) \leq L(f, P') \leq U(f, P') \leq U(f, P).$$

Let $\{P_n\}_{n=1}^\infty$ be a sequence of partitions such that

$$P_n \subseteq P_{n+1} \quad \text{and} \quad \|P_n\| \rightarrow 0.$$

Then

$$L(f, P_n) \uparrow \int_a^b f(t) \, dt,$$

where the limit is called the *Riemann lower integral*, and

$$U(f, P_n) \downarrow \overline{\int_a^b f(t) \, dt},$$

where the limit is called the *Riemann upper integral*.

Remark 4.1. If

$$Q_n \subseteq Q_{n+1}, \quad \|Q_n\| \rightarrow 0$$

is another sequence of partitions, then both the lower and upper integrals obtained above are the same. Indeed, one may take common refinements of P_n and Q_n , whose mesh also tends to 0. Thus the Riemann lower and upper integrals are independent of the particular nested sequence of partitions used.

Definition 4.2. A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is called *Riemann integrable* iff

$$\int_a^b f(t) dt = \overline{\int_a^b f(t) dt}.$$

Their common value is called the *Riemann integral* of f , and is denoted by

$$\Re \int_a^b f(t) dt.$$

Note. In these set of notes, for the sake of clarity we write Riemann integrals as $\Re \int$ and Lebesgue integrals as $\int$

Example 4.3. Consider $f = \mathbf{1}_{\mathbb{Q} \cap [a, b]}$. Then every subinterval contains both rational and irrational numbers, and hence

$$\int_a^b f(t) dt = 0, \quad \overline{\int_a^b f(t) dt} = b - a.$$

Therefore f is not Riemann integrable.

Remark 4.4. Consider $([a, b], \mathcal{B}_{[a, b]}, \lambda)$ where λ denotes Lebesgue measure. For a partition $P_n = \{x_0, \dots, x_n\}$ define the step functions

$$U_n(f, x) = \begin{cases} \sup_{x_i \leq x \leq x_{i+1}} f(x), & x_i < x \leq x_{i+1}, \\ 0, & x = a, \end{cases}$$

$$L_n(f, x) = \begin{cases} \inf_{x_i \leq x \leq x_{i+1}} f(x), & x_i < x \leq x_{i+1}, \\ 0, & x = a. \end{cases}$$

Then

$$U(f, P_n) = \int_{[a, b]} U_n d\lambda, \quad L(f, P_n) = \int_{[a, b]} L_n d\lambda.$$

Moreover, $U_n(x) \downarrow U(x)$ and $L_n(x) \uparrow L(x)$ and both U_n, L_n, U, L are measurable. Hence, by the Monotone Convergence Theorem,

$$\int_{[a, b]} L_n d\lambda \uparrow \int_{[a, b]} L d\lambda, \quad \text{and} \quad \int_{[a, b]} U_n d\lambda \downarrow \int_{[a, b]} U d\lambda.$$

If f is Riemann integrable, then

$$R \int_a^b f(t) dt = \int_a^b f(t) dt = \overline{\int_a^b f(t) dt}.$$

Therefore

$$\int_{[a, b]} L d\lambda = \int_{[a, b]} U d\lambda = R \int_a^b f(t) dt.$$

Since $L(x) \leq f(x) \leq U(x)$, we have

$$\int_{[a, b]} (U - L) d\lambda = 0, \implies \boxed{L = U \quad \text{a.e. } [\lambda]}$$

Therefore f is almost surely measurable and

$$\int_{[a, b]} f d\lambda = \Re \int_a^b f(t) dt.$$

Theorem 4.5 (Lebesgue's Criterion for Riemann Integrability). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded.

- (a) If f is Riemann integrable, then f is almost surely measurable and

$$\int_{[a,b]} f \, d\lambda = R \int_a^b f(t) \, dt.$$

- (b) f is Riemann integrable **if and only** if the set of points of discontinuity of f has Lebesgue measure zero.

Remark 4.6. If $x \notin \bigcup_{n=1}^{\infty} P_n$ then the intervals containing x shrink to x . Hence $L(x) \leq f(x) \leq U(x)$. In particular, if $L(x) = U(x)$ then f is continuous at x .

5 Non-Classical Set-Up

The classical Riemann integral assumes:

1. the domain is a compact interval $[a, b]$
2. f is bounded;
3. hence the function and the interval both satisfy the classical assumptions.

We now consider situations where one or both of these assumptions fail.

Example 5.1. Let

$$f(x) = \frac{(-1)^{n-1}}{n}, \quad n-1 \leq x \leq n, \quad n \geq 1.$$

Then f is defined on $[0, \infty)$ and

$$\Re \int_0^N f(t) \, dt = \sum_{n=1}^N \frac{(-1)^{n-1}}{n}.$$

Hence

$$\lim_{N \rightarrow \infty} \Re \int_0^N f(t) \, dt = \ln 2.$$

Thus the improper Riemann integral exists. ($\Re \int_0^\infty f(t) \, dt = \ln 2$). However,

$$f^+(x) = \frac{1}{2n+1}, \quad f^-(x) = \frac{1}{2n},$$

on the corresponding intervals. Hence

$$\int_0^\infty f^+ \, d\lambda = \int_0^\infty f^- \, d\lambda = \infty.$$

Therefore f is not Lebesgue integrable.

Remark 5.2. The improper Riemann integral can exist even though the Lebesgue integral does not exist. The issue is that the positive and negative parts are both infinite.

Example 5.3. Similarly,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} \, dx$$

can be considered as an improper Riemann integral, and we obtain,

$$\Re \int_{-\infty}^{\infty} \frac{\sin x}{x} \, dx = \frac{\pi}{2}$$

under the usual symmetric/improper interpretation.

Example 5.4. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x/(1 + x^2)$. Then,

$$\lim_{M, N \rightarrow \infty} \Re \int_{-N}^M f(t) \, dt$$

does not exist.

Indeed,

$$\Re \int_{-N}^M \frac{t}{1 + t^2} \, dt = \frac{1}{2} \log \left(\frac{1 + M^2}{1 + N^2} \right),$$

whose limit depends on how $M, N \rightarrow \infty$.

However the symmetric limit exists as,

$$\lim_{N \rightarrow \infty} \Re \int_{-N}^N f(t) \, dt = 0.$$

This is called the *Cauchy principal value*. This is often useful in physics for example in problems of calculating moments of inertia.

Example 5.5. Consider

$$f(x) = \frac{1}{\sqrt{x(1-x)}}, \quad 0 < x < 1.$$

Then f is unbounded near 0 and 1, so the classical Riemann integral does not apply directly.

Nevertheless, the improper Riemann integral exists, and the corresponding Lebesgue integral also exists.